

Student Satisfaction Through Decision Intelligence

Executive Business Report

AWS QuickSight Business Intelligence Case Study

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Industry: Higher Education

Domain: Business Intelligence | Decision Intelligence | Executive Analytics

Technology: Amazon QuickSight

1. Executive Summary

Higher education institutions must continuously balance student experience, teaching performance, institutional demand, and financial sustainability. The challenge is not simply to understand what has happened, but to determine where leadership should focus attention and resources to improve outcomes without unnecessarily increasing expenditure.

This case study applies Business Intelligence and Decision Intelligence techniques to student enrollment, professor evaluation, course evaluation, and course cost data from a regional community college.

Within this analysis, **student satisfaction is assessed primarily through the professor and course evaluation scores available in the dataset.** The solution therefore focuses on identifying patterns associated with stronger evaluation outcomes and examining those outcomes alongside course cost.

The analytical environment was developed using **Amazon QuickSight**, incorporating data preparation and calculated fields, KPI analysis, interactive dashboards, natural-language analytics through QuickSight Topics, scenario-based analysis, and Data Storytelling. The original project included dataset configuration, analysis visuals, dashboard development, verified natural-language answers, a structured scenario thread, and a complete Data Story.

The analysis produced a decision framework centred on three strategic opportunities:

1. **Identify and investigate practices associated with consistently high-performing professors.**
2. **Prioritise courses demonstrating strong evaluation outcomes alongside relatively low delivery costs.**

3. **Use observed improvements, particularly the 2022 performance changes in Accounting and Statistics, as opportunities for institutional learning and replication.**

The resulting recommendation is not simply to spend more to improve student outcomes. Instead, leadership should use performance evidence to identify successful practices, assess their transferability, and focus resources on areas demonstrating strong educational value relative to cost.

2. Business Challenge

Higher education institutions operate under competing pressures.

Leadership must improve teaching quality and student experience while maintaining financial discipline and making effective use of limited institutional resources.

In the absence of an integrated analytical framework, performance information can remain fragmented across different dimensions:

- professor performance;
- course evaluation outcomes;
- enrollment patterns;
- student segments;
- course delivery costs.

Historical reporting can describe these measures individually, but it does not necessarily answer the executive questions that matter most:

Where are the strongest performance patterns?

Which improvements appear to offer the greatest opportunity for replication?

Where is strong student evaluation performance being achieved efficiently?

How can leadership improve outcomes without assuming that additional expenditure is the only solution?

The original project framed the central business challenge as improving professor evaluation performance and student satisfaction while maintaining cost efficiency.

The Business Intelligence solution was therefore designed not merely to report performance, but to provide a structured basis for evaluating these trade-offs.

3. Decision to Be Made

The central executive decision was:

How can the institution improve professor performance and student satisfaction while avoiding an increase in cost per course?

This question deliberately creates a tension between two objectives:

Educational performance

versus

Financial efficiency

Rather than assuming that improvement requires additional spending, the analysis investigates whether existing high-performing examples can provide a more efficient path to improvement.

This reframes the BI problem from:

“What happened?”

to:

“Where should leadership focus its attention, and which observed successes are worth investigating and replicating?”

That shift from descriptive reporting toward decision support is the central Decision Intelligence principle applied in this case study.

4. Strategic Objectives

The solution was designed to support six strategic objectives.

4.1 Improve Indicators of Student Satisfaction

Identify patterns associated with stronger professor and course evaluation outcomes.

4.2 Strengthen Evidence-Based Decision-Making

Provide leadership with accessible performance information through executive dashboards and analytical exploration.

4.3 Improve Resource Allocation

Identify courses demonstrating a strong combination of evaluation performance and cost efficiency.

4.4 Identify Replication Opportunities

Identify consistently high-performing professors, courses, and observed improvement patterns that warrant further investigation.

4.5 Strengthen Performance Governance

Create a repeatable framework for monitoring evaluation performance, enrollment, and cost-related KPIs.

4.6 Support Strategic Planning

Transform operational data into structured evidence that can inform institutional improvement decisions.

5. Business Context

Higher education institutions must balance student experience, teaching quality, institutional demand, and financial sustainability while making effective use of available resources.

Business Intelligence can support this balance by connecting academic and operational data into a coherent performance view.

In this case, the analytical environment brings together student enrollment, student type, professor evaluation, course evaluation, and course cost perspectives. The objective is not to claim that these variables independently explain student satisfaction, but to identify observable performance patterns that can inform further management investigation.

This distinction is important.

The BI solution identifies evidence and opportunities for decision-making; it does not claim to establish causal relationships that the available dataset cannot prove.

6. Analytical Approach

A structured BI methodology was used to move from operational data toward executive decision support.

The workflow incorporated five major stages.

6.1 Data Preparation & Modelling

The analytical dataset was configured within Amazon QuickSight and enriched through calculated fields.

One example was the creation of a **Student Type** field that classified students according to age:

```
ifelse({Age} < 30, "Youth", "Adult Continuing Education")
```

The original project documentation records dataset field configuration, refresh scheduling, and calculated-field development as part of the BI workflow.

6.2 Performance Measurement

The analysis examined several dimensions of institutional performance, including:

- student enrollment;
- student type;
- professor evaluation scores;
- course evaluation scores;
- average course cost;
- performance trends over time.

These measures provided the analytical foundation for comparing academic outcomes and operational efficiency.

The original analysis included visuals covering student majors by year, student type proportions, professor evaluation scores, course evaluation scores, professor-level course costs, and course-level course costs.

6.3 Executive Visualisation

The analytical outputs were translated into executive dashboards designed to make performance patterns easier to identify and compare.

The dashboard environment consolidated multiple perspectives rather than forcing leadership to interpret isolated charts.

6.4 Scenario-Based Decision Analysis

The analysis then moved beyond visual exploration into structured questioning.

The scenario began with broad questions about professor performance and progressively narrowed toward:

- year-over-year improvement;
- Accounting and Statistics;
- high-evaluation, lower-cost courses;
- Data Visualization as a cost-effective example;
- the final institutional recommendation.

The original analytical thread explicitly followed this progression from broad professor-level performance to course-level opportunities and ultimately to strategic recommendations.

6.5 Business Storytelling

The final analytical findings were translated into a Data Story combining visuals, narrative interpretation, and recommendations.

This created a communication layer between the analytical work and executive decision-making.

7. BI Solution

The solution was implemented within Amazon QuickSight as an integrated Business Intelligence environment.

The architecture included:

Data Preparation → Analytical Model → KPI Analysis → Natural-Language Analytics → Scenario Analysis → Executive Dashboards → Data Story

The original project included QuickSight datasets, analyses, dashboards, Topics, verified answers, scenarios, and a Data Story.

8. Executive Dashboard Solution

The dashboard environment was structured around four complementary analytical perspectives.

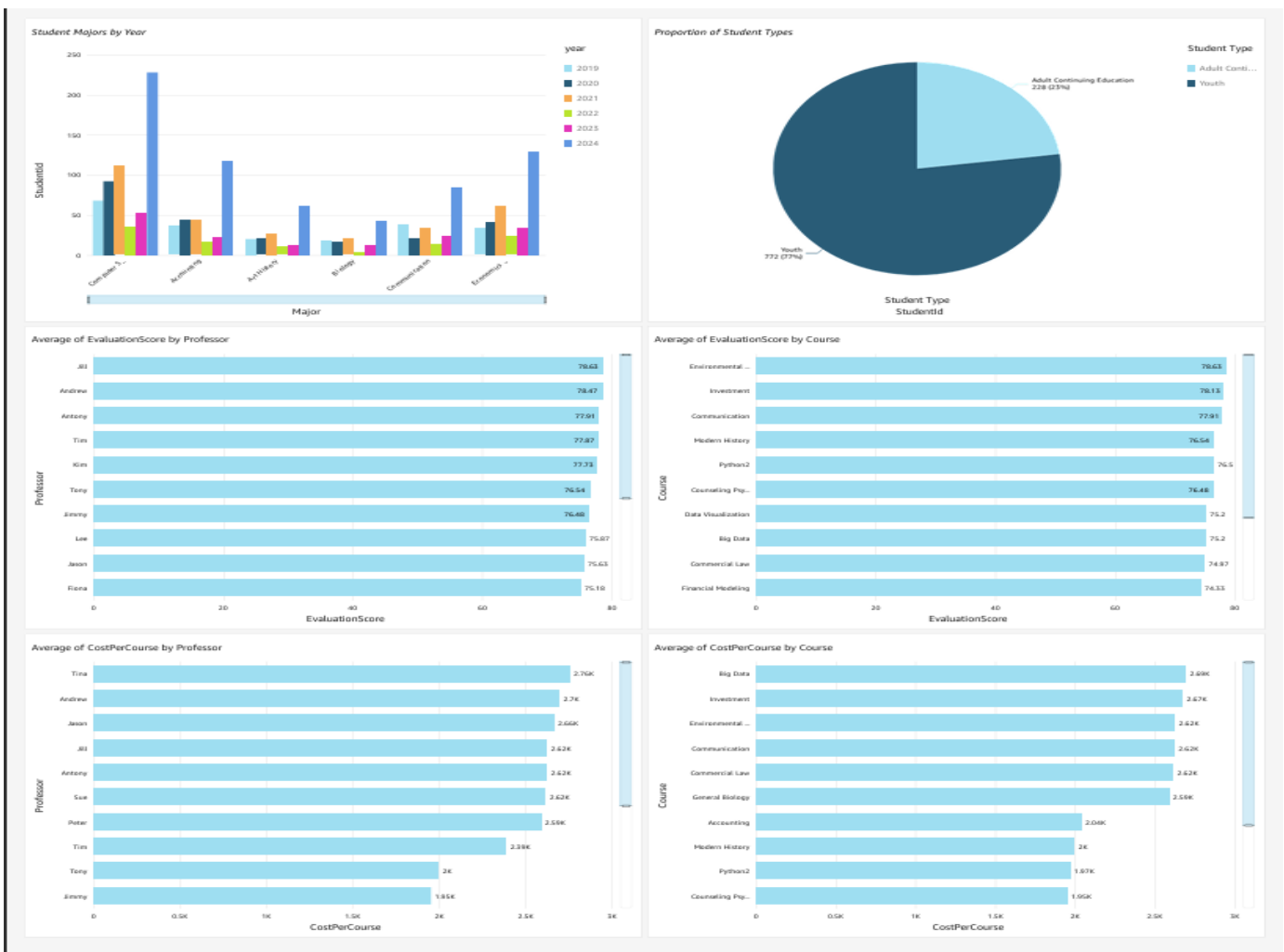
Dashboard 1 — Executive Performance Overview

Purpose

Provide leadership with a consolidated view of enrollment, professor performance, course evaluations, and cost efficiency.

Decision Value

Enable rapid identification of performance trends, gaps, and areas requiring deeper investigation.



Dashboard 2 — Professor Performance Analysis

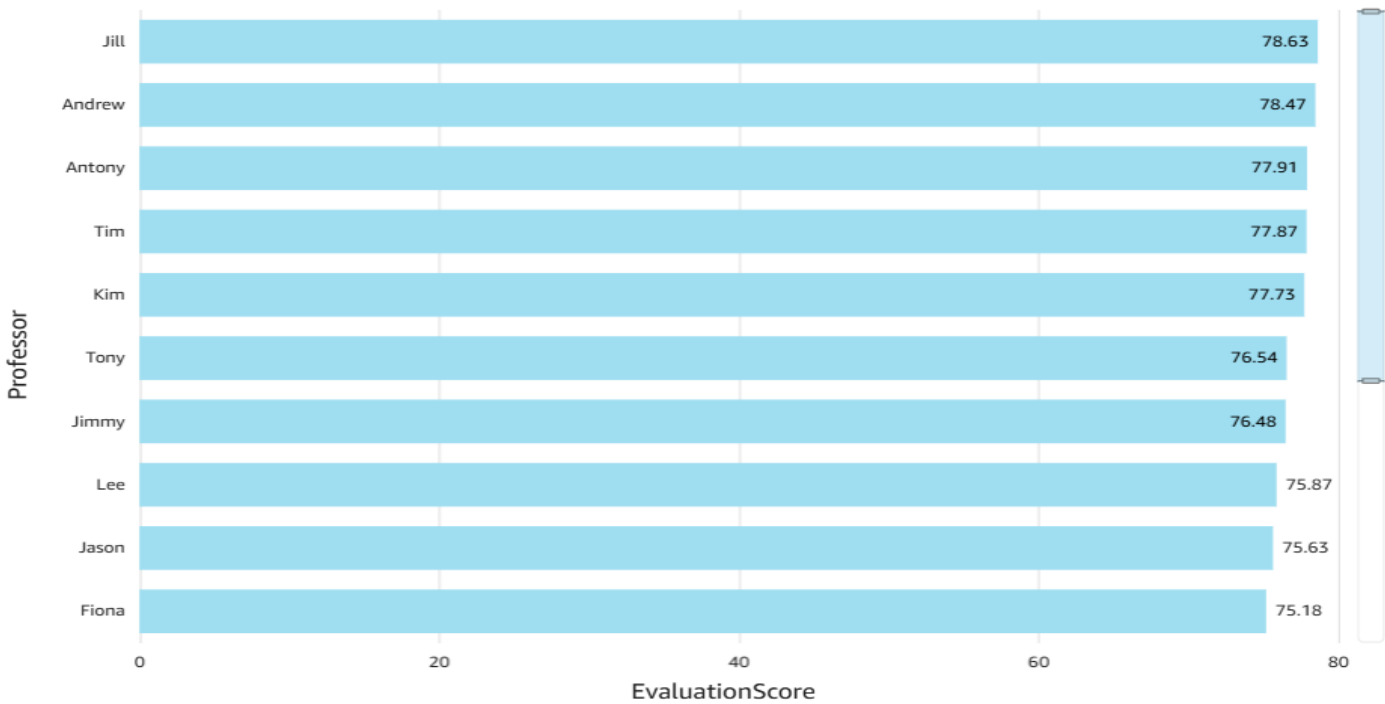
Purpose

Compare professor evaluation performance and identify consistently high-performing faculty.

Decision Value

Provide an evidence base for investigating successful teaching practices and potential faculty-development opportunities.

Average of EvaluationScore by Professor



Dashboard 3 — Course Performance & Cost Efficiency

This view combines two complementary perspectives:

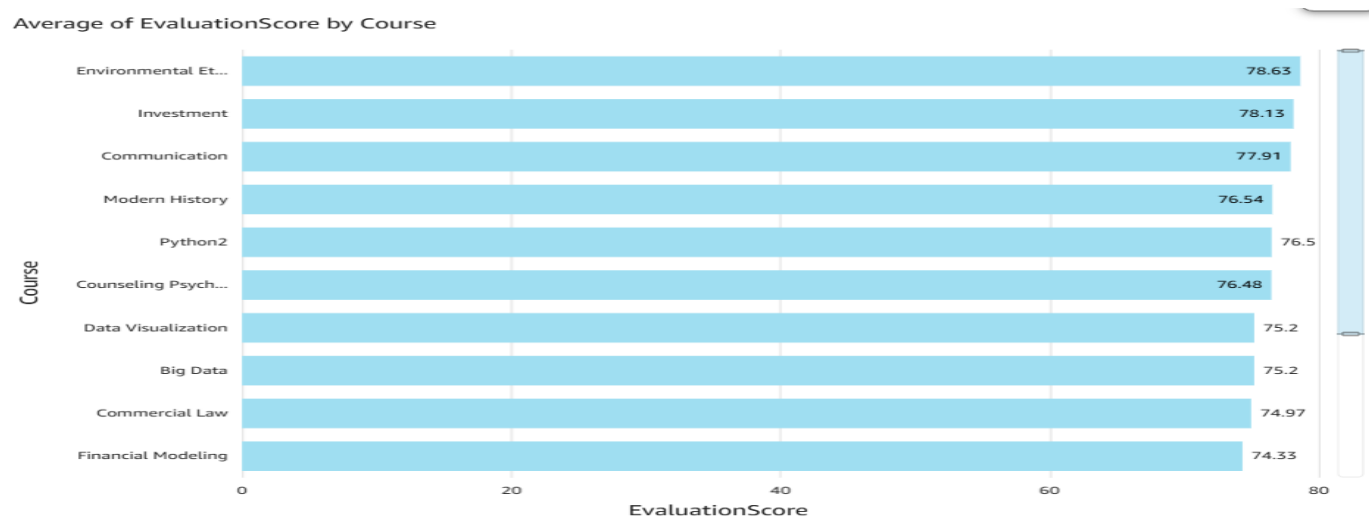
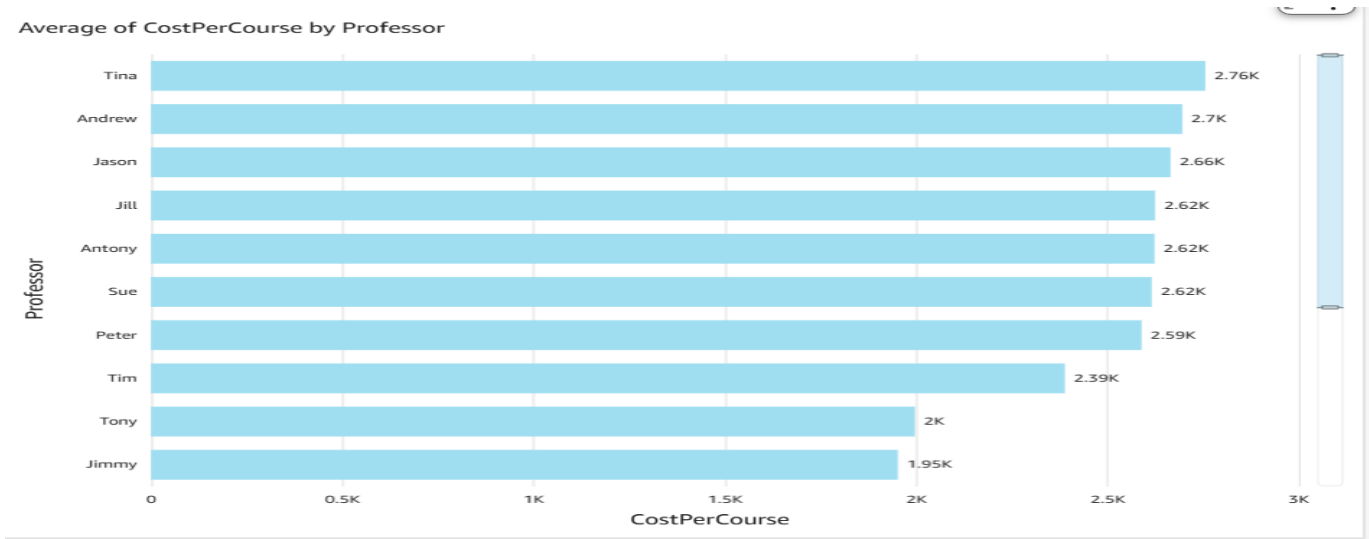
- Course Cost Efficiency
- Course Evaluation Performance

Purpose

Examine course outcomes alongside delivery costs.

Decision Value

Help leadership identify courses where strong evaluation performance is being achieved without correspondingly high delivery costs.



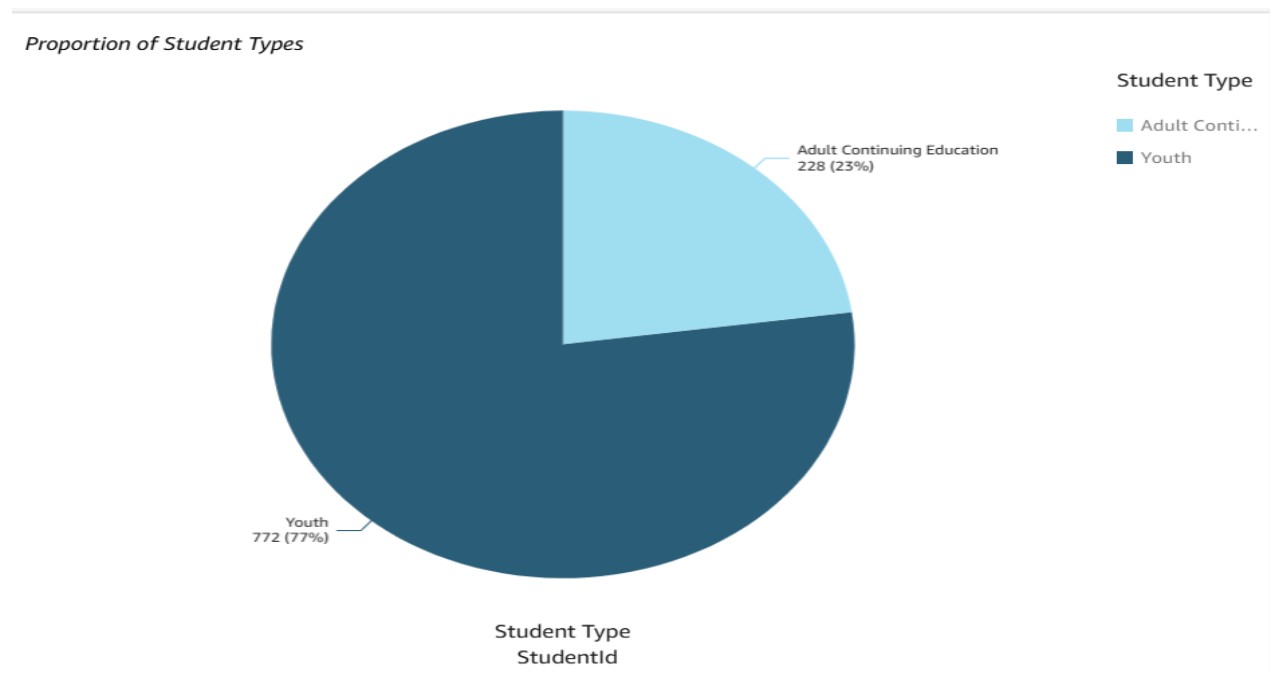
Dashboard 4 — Student Demographics & Enrollment Trends

Purpose

Examine enrollment patterns and student demographics.

Decision Value

Provide visibility into student populations and participation trends that can inform institutional planning.



9. Key Findings

The analysis produced five principal findings.

Finding 1 — High-Performing Professors Represent a Replication Opportunity

The analysis identified professors with strong evaluation performance.

The strategic value is not simply recognising individual high performers. The more important question is whether the practices associated with their performance can be investigated and potentially transferred to other teaching environments.

This creates a potential **replication strategy** rather than a one-time recognition exercise.

Finding 2 — 2022 Performance Improvement Provides a Replication Opportunity

Evaluation performance improved in 2022 compared with previous periods.

This creates an opportunity for leadership to investigate what changed during the period and determine which practices or course-level improvements may be worth sustaining.

The original scenario specifically investigated the factors contributing to higher evaluation scores in 2022.

Importantly, the analysis identifies an opportunity for investigation and replication; it does not establish that the observed changes were caused by one specific intervention.

Finding 3 — Course-Level Improvements Can Create Institutional-Level Impact

Accounting and Statistics were highlighted as important contributors to the improvement observed in 2022.

This suggests that targeted course-level improvements can have broader institutional relevance.

From a management perspective, this creates a useful question:

Which successful course-level practices are sufficiently transferable to justify replication elsewhere?

The scenario analysis explicitly positioned Accounting and Statistics as potential leverage points for institutional improvement.

Finding 4 — Strong Evaluation Outcomes Can Coexist With Lower Cost

The analysis identified courses that combined relatively strong evaluation performance with relatively low delivery costs.

This challenges the assumption that improved student outcomes necessarily require proportional increases in expenditure.

The implication is not that cost should become the dominant decision criterion. Rather, cost and performance should be evaluated together.

Finding 5 — Data Visualization Demonstrates Potential for Cost-Effective Excellence

Data Visualization emerged as an example of strong evaluation performance alongside relatively low course cost.

This makes it a useful candidate for deeper investigation.

Leadership could examine the characteristics and teaching practices associated with this performance and determine whether any are transferable to other courses.

The original scenario specifically used Data Visualization to investigate how strong evaluation performance could coexist with low cost.

10. Executive Interpretation

Taken together, the findings point toward a common strategic theme:

The strongest opportunity may not be simply increasing expenditure, but identifying, understanding, and replicating practices already associated with strong outcomes.

The analytical framework enables leadership to distinguish between:

- high performance;
- cost efficiency;
- improvement over time;
- potential replication opportunities.

This supports more targeted management attention and reduces the risk of making resource-allocation decisions based solely on isolated performance indicators.

11. Strategic Recommendations

Recommendation 1 — Replicate High-Performing Teaching Practices

Identify the teaching practices associated with consistently high professor evaluation performance and investigate whether they can be shared or replicated across departments.

Executive rationale: Scale observed sources of strong evaluation performance rather than treating high performance as isolated success.

Recommendation 2 — Prioritise High-Value, Cost-Efficient Courses

Evaluate course evaluation outcomes alongside delivery costs to identify courses demonstrating a strong combination of educational performance and financial efficiency.

Executive rationale: Focus management attention and resources on areas demonstrating measurable value.

Recommendation 3 — Use 2022 as a Performance Benchmark

Investigate the changes associated with improved evaluation performance during 2022, with particular attention to Accounting and Statistics.

Executive rationale: Convert observed improvement into institutional learning rather than allowing successful performance to remain isolated.

Recommendation 4 — Investigate Data Visualization as a Cost-Effective Model

Conduct a deeper review of the practices associated with Data Visualization's strong evaluation performance and relatively low cost.

Executive rationale: Determine whether the conditions behind this performance can inform broader efficiency strategies.

Recommendation 5 — Institutionalise Executive Performance Monitoring

Maintain a repeatable BI framework covering:

- professor evaluations;
- course evaluations;
- enrollment;
- student segments;
- course cost;
- performance trends.

Executive rationale: Move from periodic retrospective reporting toward continuous evidence-based performance management.

12. Strategic Priority

Improve student satisfaction through replication, optimisation, and continuous measurement—not simply through increased spending.

By combining performance evidence with cost-efficiency analysis, leadership can focus limited resources on interventions with the strongest potential institutional value.

13. Business Value

Because this project represents an analytical case study rather than a post-implementation client engagement, the business value described here refers to the **decision-support capabilities enabled by the solution**, rather than measured organisational outcomes.

The solution enables leadership to:

Improve Decision Visibility

Provide a consolidated view of academic and operational performance.

Support Evidence-Based Resource Allocation

Compare course evaluation outcomes with delivery costs when evaluating resource priorities.

Identify Replication Opportunities

Use performance evidence to identify high-performing professors and courses worthy of deeper investigation.

Strengthen Continuous Performance Management

Establish a repeatable framework for monitoring performance over time.

Improve Strategic Planning

Connect operational evidence with institutional objectives and strategic priorities.

Executive Impact Statement

The primary value of the solution is its ability to convert fragmented academic and operational data into a structured decision-support framework, enabling leadership to evaluate performance, identify cost-effective opportunities, and make more targeted decisions about institutional improvement.

14. Technical Architecture

Cloud BI Environment

Amazon QuickSight served as the core Business Intelligence environment.

The solution incorporated:

- dataset configuration;
- calculated fields;
- analytical visuals;
- dashboards;
- QuickSight Topics;
- verified answers;
- scenario analysis;
- Data Storytelling.

The original project documentation confirms the use of QuickSight for the complete workflow from dataset preparation through scenario analysis and Data Story development.

Data Preparation & Modelling

The dataset was configured within QuickSight and enriched through calculated fields.

The Student Type field provided a segmentation mechanism distinguishing Youth from Adult Continuing Education students.

Dataset Management

A dataset refresh schedule was configured to support data currency and reliability within the analytical environment.

Analytical Layer

Multiple analyses examined:

- enrollment;

- student type;
- professor evaluations;
- course evaluations;
- course costs.

These analyses provided the evidence base for dashboard development and scenario analysis.

Natural-Language Analytics

A QuickSight Topic was configured with named entities and verified answers to support natural-language exploration of student, professor, and course information.

The original deliverables explicitly included named entities and verified answers as part of the QuickSight implementation.

Scenario Analysis

The scenario analysis used a structured sequence of questions to move from broad performance analysis toward targeted course-level investigation and strategic recommendations.

This created a reasoning layer between raw analytical outputs and executive action.

Dashboard & Data Story

The analytical outputs were consolidated into executive dashboards and a Data Story designed to communicate the findings to business stakeholders.

15. Technology Stack

Amazon QuickSight

Business Intelligence

Data Modelling & Transformation

Calculated Fields

KPI Analysis

Dashboard Development

Natural-Language Analytics

Scenario Analysis

Data Storytelling

16. Lessons & Future Enhancements

The analysis also identified several opportunities for extending the solution into a more mature institutional decision-support environment.

16.1 Expand Performance Monitoring

Introduce additional measures of student experience, faculty performance, enrollment behaviour, and course effectiveness.

16.2 Strengthen Longitudinal Analysis

Extend the analytical period to distinguish sustained performance patterns from short-term changes.

16.3 Establish Comparative Benchmarks

Create consistent benchmarks for professors and courses to support performance comparison and improvement monitoring.

16.4 Expand Scenario Analysis

Develop additional scenarios examining trade-offs among:

- student evaluation outcomes;

- course cost;
 - enrollment;
 - resource allocation.
-

16.5 Integrate Additional Data Sources

Where available, integrate additional institutional data such as:

- retention;
- attendance;
- course completion;
- other student outcome measures.

This would provide a broader view of institutional performance.

16.6 Strengthen Executive Decision Support

Evolve the environment toward a more comprehensive decision-support platform combining:

KPI Monitoring → Comparative Analysis → Scenario Exploration → Automated Insight Generation

These future enhancements are consistent with the original project's documented opportunities for expanding longitudinal analysis, benchmarking, scenario analysis, data integration, and executive decision support.

17. Professional Reflection

This project reinforces a fundamental principle of Business Intelligence:

The value of analytics is not determined by the number of dashboards or visualisations produced, but by the quality of the decisions those insights enable.

A mature BI environment should progressively evolve from historical reporting toward:

- performance diagnosis;
- opportunity identification;
- scenario evaluation;

- decision support;
- continuous performance management.

The project demonstrates that transition by combining analytical modelling, visualisation, structured questioning, and executive storytelling within a single workflow.

18. Conclusion

This case study demonstrates how Amazon QuickSight can be used to transform academic and operational data into structured decision support.

The analytical framework connected professor evaluation performance, course evaluation outcomes, enrollment patterns, student segmentation, and course cost efficiency.

The resulting insights point toward a targeted strategy:

1. identify high-performing practices;
2. investigate successful improvement patterns;
3. identify high-value, cost-efficient courses;
4. evaluate opportunities for replication;
5. continuously monitor performance through executive BI.

The central strategic lesson is straightforward:

Improvement does not necessarily begin with increased spending. It can begin with identifying where the organisation is already succeeding, understanding why, and determining whether those practices can be scaled.

That is where Business Intelligence becomes more than reporting—it becomes a mechanism for improving the quality, consistency, and discipline of organisational decision-making.

Appendix A — Supporting Evidence

The complete project evidence is maintained separately from this executive report.

Supporting materials include:

- Original project documentation;
- Executive dashboards;
- Scenario analysis;

- Data Story;
- QuickSight analytical deliverables;
- Project repository materials.

The original project included the complete BI workflow and documented deliverables across datasets, analyses, dashboards, Topics, scenarios, and Data Storytelling.

Appendix B — Project Deliverables

Deliverable	Purpose
Dataset	Analytical foundation
Calculated Fields	Data enrichment and segmentation
Analyses	Detailed performance exploration
Dashboard	Executive performance monitoring
QuickSight Topic	Natural-language analytics
Verified Answers	Validated analytical responses
Scenario Analysis	Structured decision reasoning
Data Story	Narrative communication of findings
Executive Business Report	Senior-level business communication
Original Project Documentation	Detailed implementation evidence

Appendix C — Recommended Evidence Map

The executive report should be read together with the supporting GitHub repository:

Executive Business Report

→ Senior-level business narrative and decision framework

Dashboard Evidence

→ Visual proof of the analytical solution

Scenario Analysis

→ Evidence of structured analytical reasoning

Data Story

→ Evidence of business storytelling and insight communication

Original Project Documentation

→ Detailed implementation and project-submission evidence

README

→ Technical project overview and repository entry point